Victor Lu

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PROFESSIONAL SUMMARY

Carnegie Mellon University Master of Entertainment Technology student with a background in engineering and art. Experienced in Unreal Engine and Unity in team projects and plays a role across the programming and art team. Made shaders, VFX, 3D modeling, and game programming experience in XR and video game projects.

EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Master of Entertainment Technology — Aug. 2025 – May. 2027

University of Illinois at Urbana Champaign — Champaign, IL
Bachelor of Science, Computer Engineering, Minor in Arts and Design — Aug. 2021 – May. 2025

SKILLS

Tools and Software: Unreal Engine, Unity, C++, Python, HLSL, Blender, Substance Painter, Substance Designer, Procreate
Skills and Interests: Shader Development, Tools Development, Rigging and Animation, Linear Algebra, 3D Modeling, Programming

PROJECTS

Anime Shading Model — May. 2025 – Jun. 2025

- Modified Unreal Engine shading model calculation through C++ and HLSL for cel-shading and added customizability to artists.
- Devised a shadow calculation falloff with vector math and created a shadow curve for adjusting colors and sharpness.
- Remodeled shading calculations for character's hair and face and depth occlusion for eyes to recreate anime effects.

Painterly Effect — Jul. 2025 – Aug. 2025

- Implemented anisotropic Kuwahara filter algorithm in Unreal Engine with HLSL to make realistic rendering stylized.
- Formulated stylized normal maps and color maps manually for objects to mimic strokes in oil paintings.
- Built a procedural painting-stroke normal and color maps generator in Substance Designer for distant objects.

PoseCapture — Mar. 2024 – May. 2024

- Developed a motion capture plugin in Python in Blender that captures actions from a video or webcam to humanoid models.
- Utilized the MediaPipe AI model to retrieve relative locations of each joint and convert it into rotation data for each bone.
- Improved workflow for animators in Blender for motion capture without additional expenditure.

EXPERIENCE

XR Game Engineer — Champaign, IL
Empathy Training Simulation — Oct. 2024 – May. 2025

- Optimized Unreal Engine to make Metahuman and other high-polygon count models supported by Meta Quests.
- Integrated large language model simulating patient with Unreal Engine so users can talk and get audio response.
- Attained funding for further development after pitching and demonstrating in the Empathy Training workshop and seminar.

Indie Game Developer — Champaign, IL
GameBuilders Studio — Sep. 2022 – May. 2025

- Created character and prop models and textures in Blender, Substance Designer, and Substance Painter.
- Accomplished animations and handled animation scripts to facilitate workflow for our programming team.
- Published three games in itch.io: https://gamebuilders-studio.itch.io/

Technical Artist Intern — Champaign, IL
Jump Simulation Center — Jun. 2024 – Aug. 2024

- Designed visual effects, including flesh deformation and cannula physics, in Unity for Meta Quests.
- Employed animation tools and shaders in Apple Developer Kit to accomplish a Lumbar Puncture tutorial for Vision Pro.
- Conducted human studies and confirmed the effectiveness of program for students to learn medical materials.